STM Exercise 7

Phase Diagrams

(partial solubility in the solid state)

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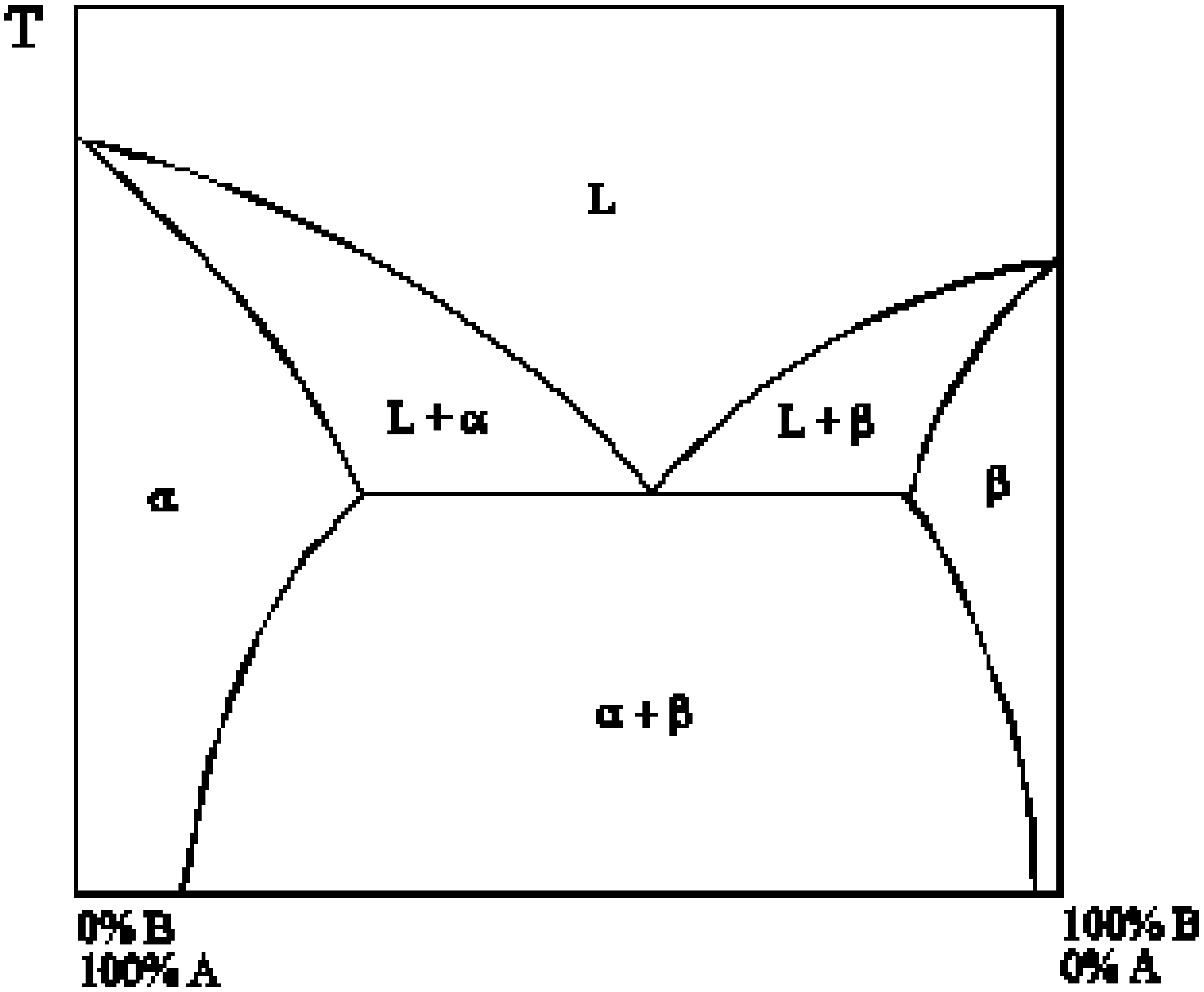
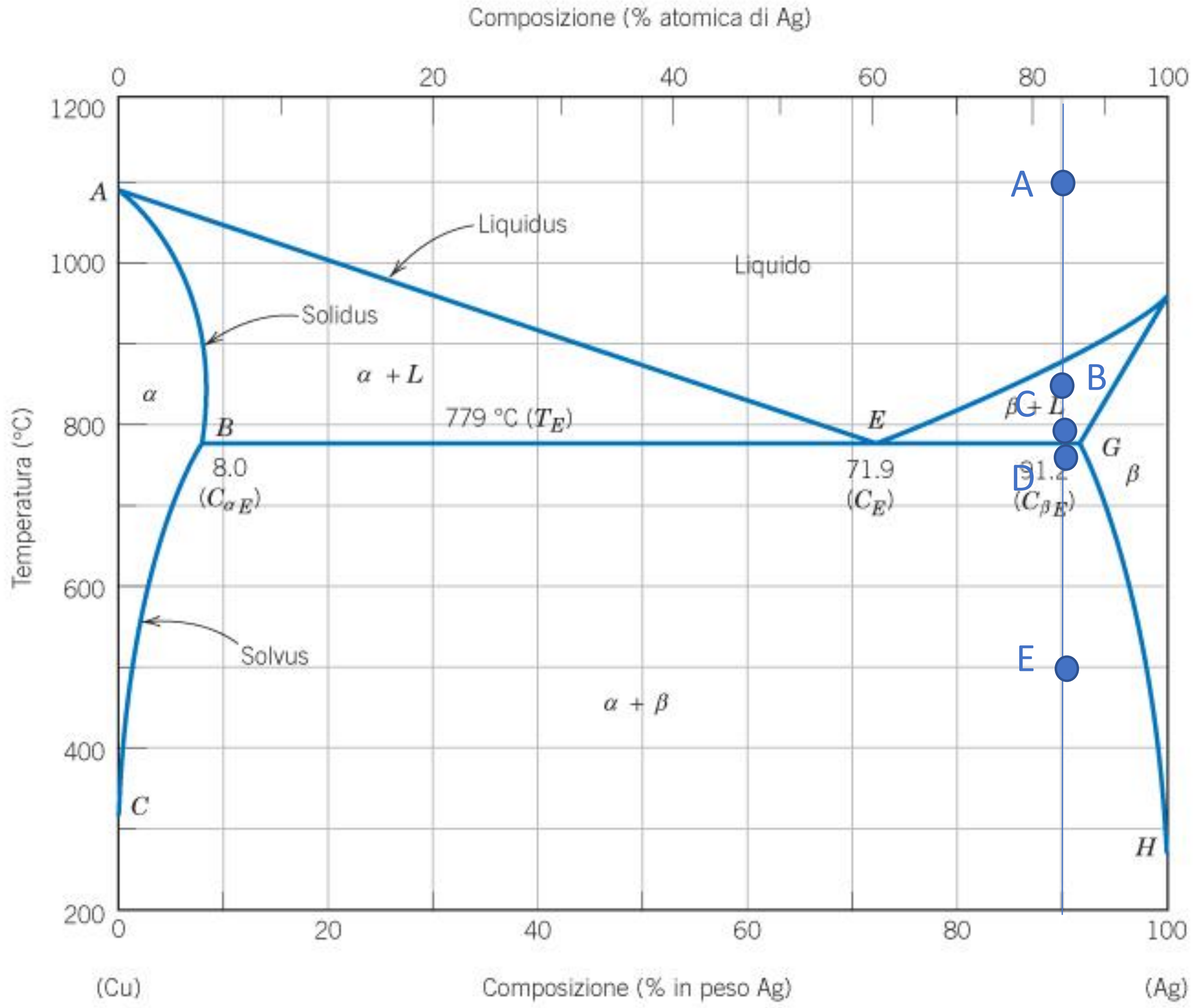
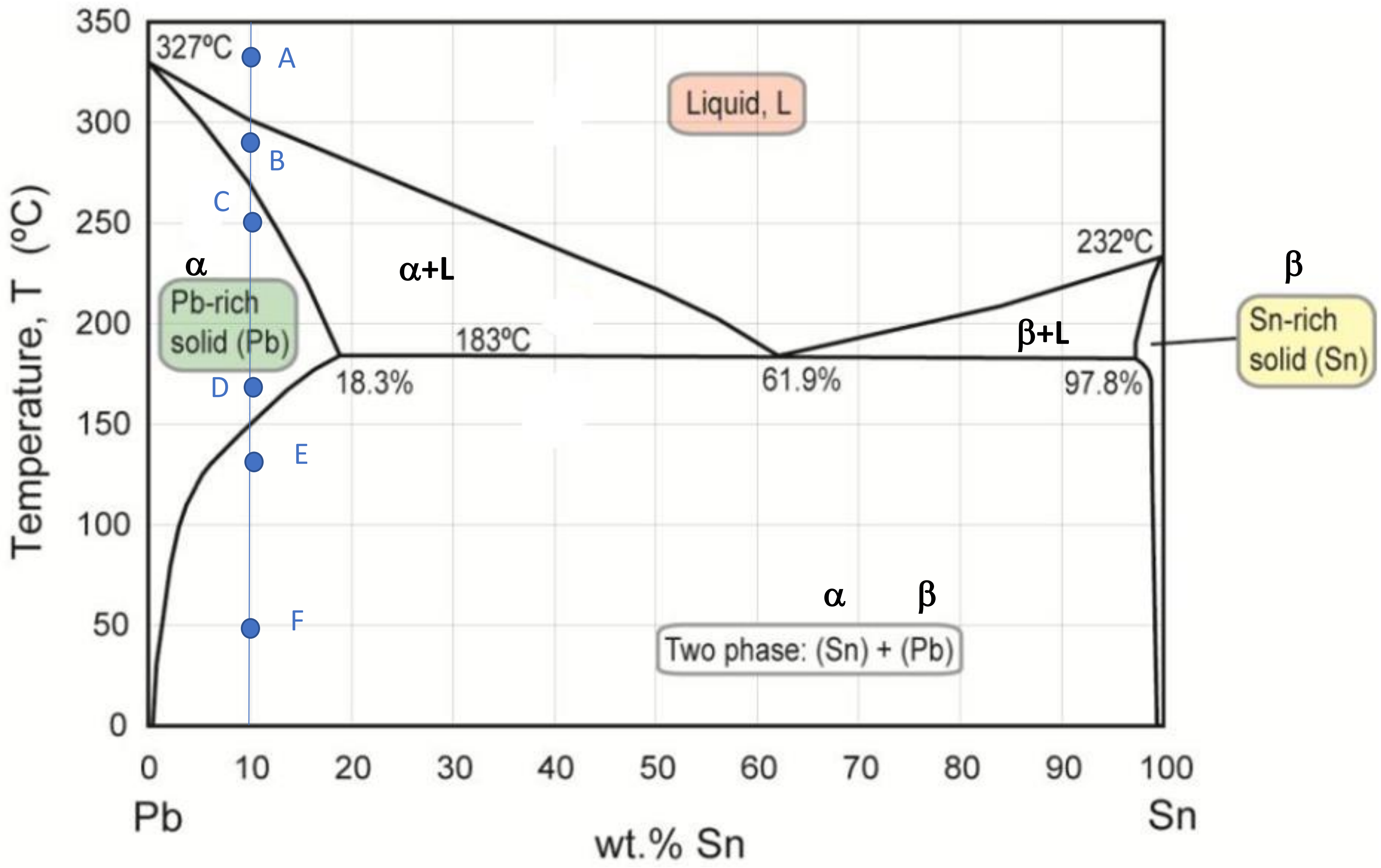
Complete solubility in the liquid state, partial solubility in the solid state



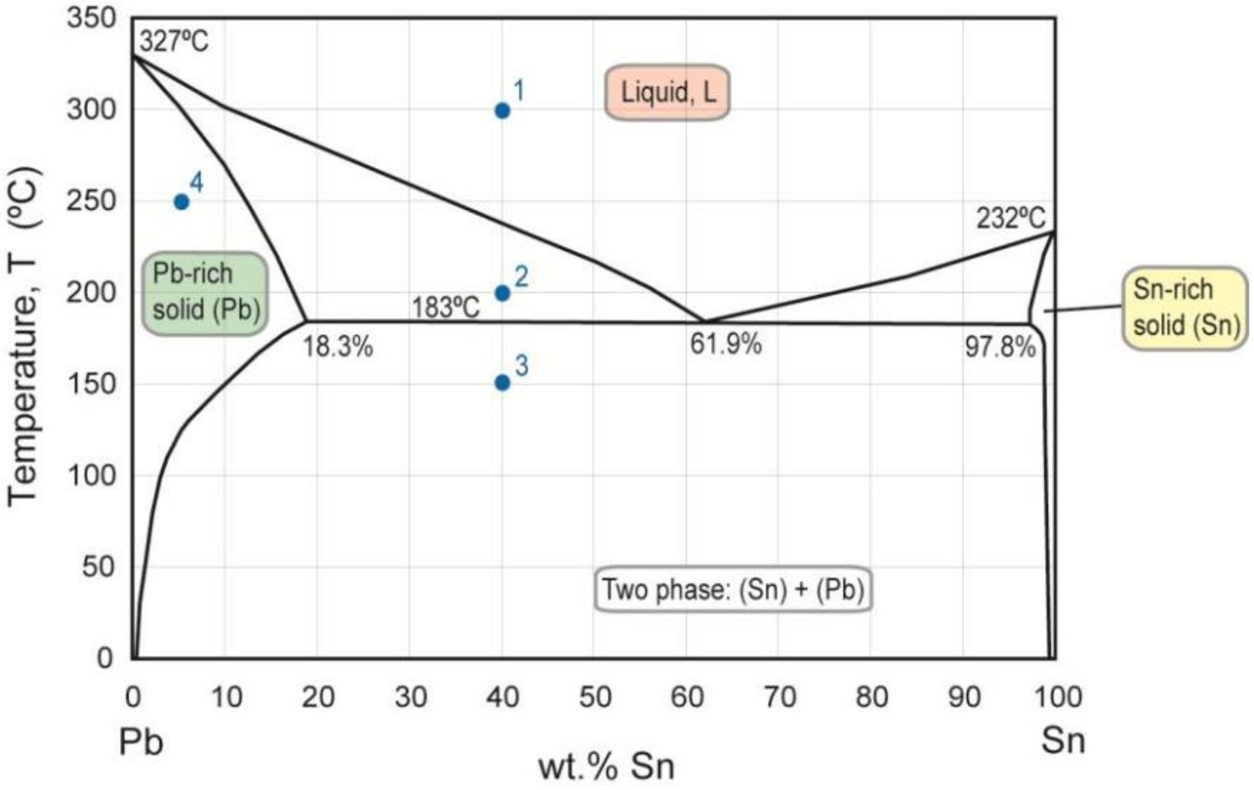
Exercise 1: determine the phase present, their composition and relative amounts in the points A, B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



Exercise 2: determine the phase present, their composition and relative amounts in the points A, B, C, D, E and F indicated in the phase diagram. Schematize the microstructure in each point.



E.6 Use the Pb-Sn diagram below to answer the following questions.



(a) What are the values of the state variables (composition and temperature) at constitution point 1?

(b) Mark the constitution points for Pb-60wt% Sn and Pb-20wt% Sn alloys at 250°C. What phases are present in each case?

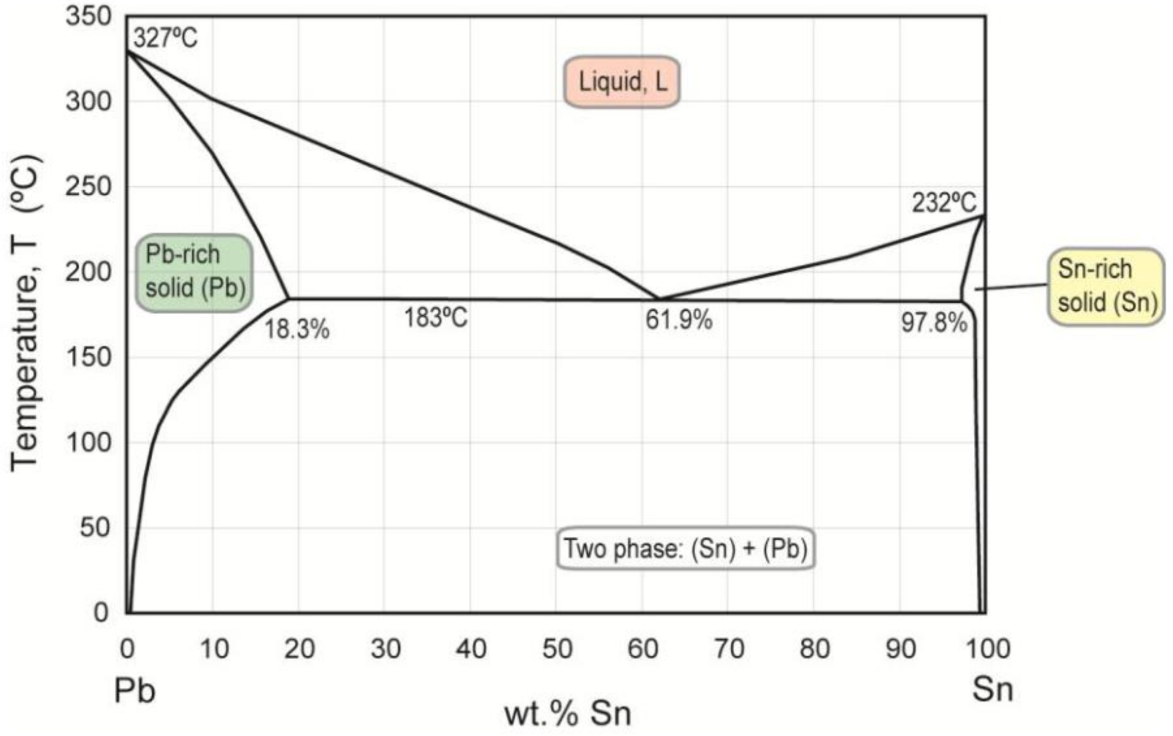
(c) The alloy at constitution point 1 is cooled very slowly to room temperature, maintaining equilibrium. At which temperatures do changes in the number or type of phases occur? What phases are present at constitution points 2 and 3?

(d) The alloy at constitution point 4 is cooled slowly to room temperature. Identify the following:

(i) the initial composition, temperature and phase(s);

(ii) the temperature at which a phase change occurs, and the final phase(s).

E.7 Use the Pb-Sn diagram below to answer the following questions.

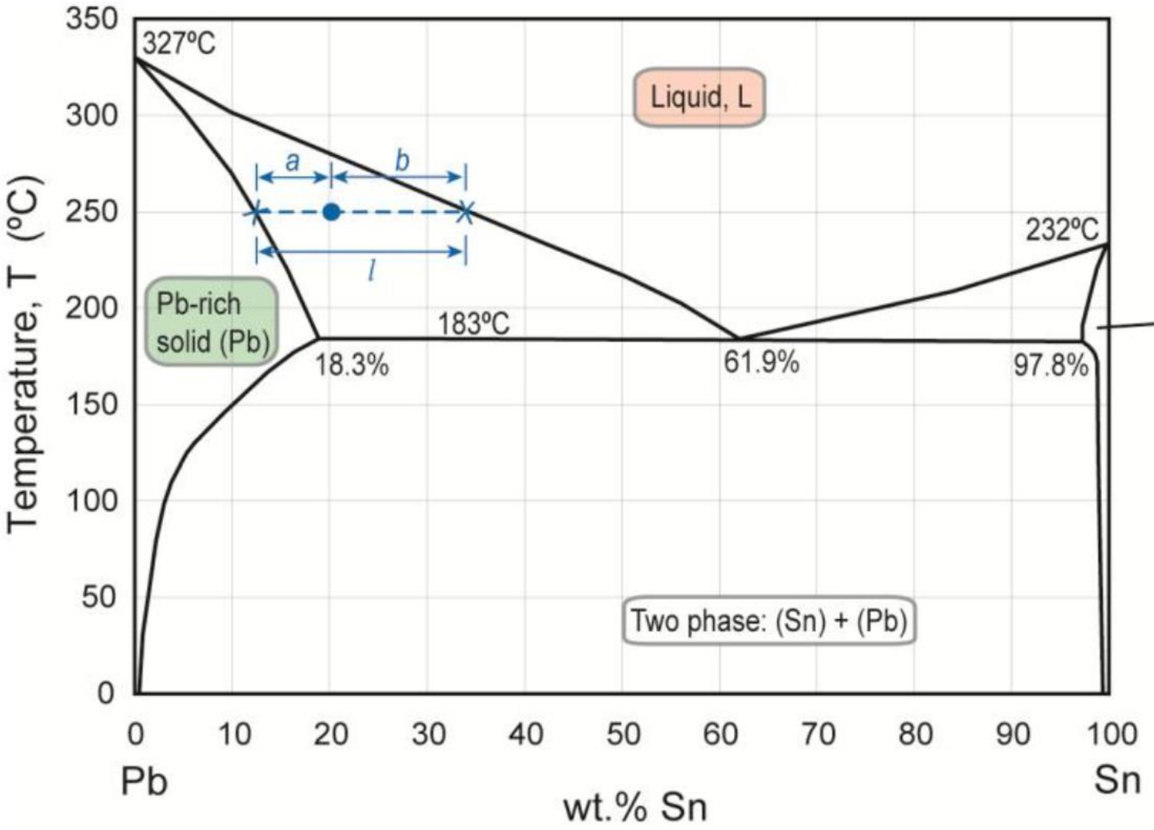


(a) The constitution point for a Pb-25wt% Sn alloy at 250 ºC lies in a two-phase field. Construct a tie-line on the figure and read off the two phases and their compositions.

(b) The alloy is slowly cooled to 200 ºC. At this temperature, identify the phases and their compositions

(c) The alloy is cooled further to 150 ºC. At this temperature, identify the phases and their compositions.

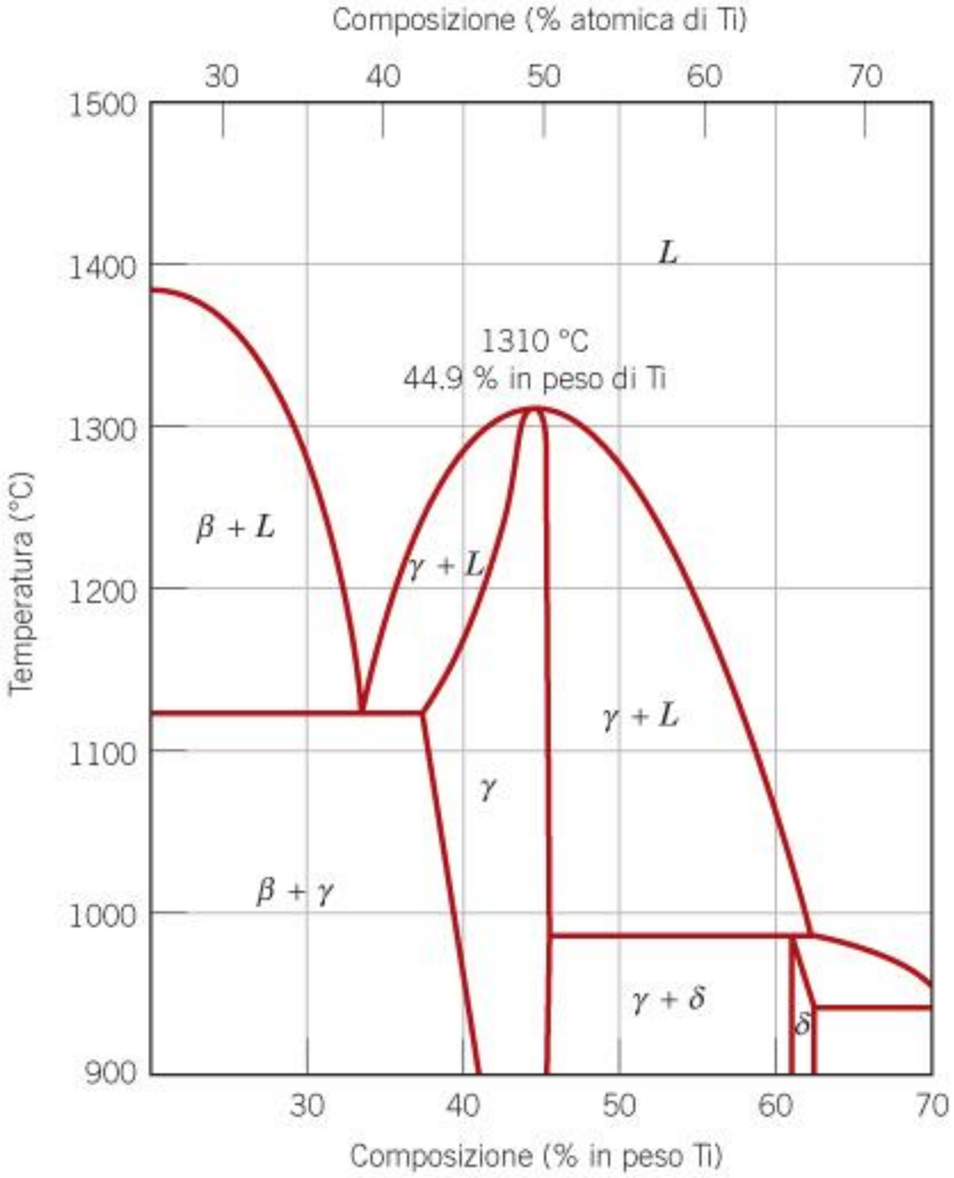
(d) Indicate with arrows on the figure the lines along which the compositions of phase 1 and phase 2 move during slow cooling from 250ºC to 200ºC.



Consider Pb-20wt% Sn at 250 ^{o}C in Figure

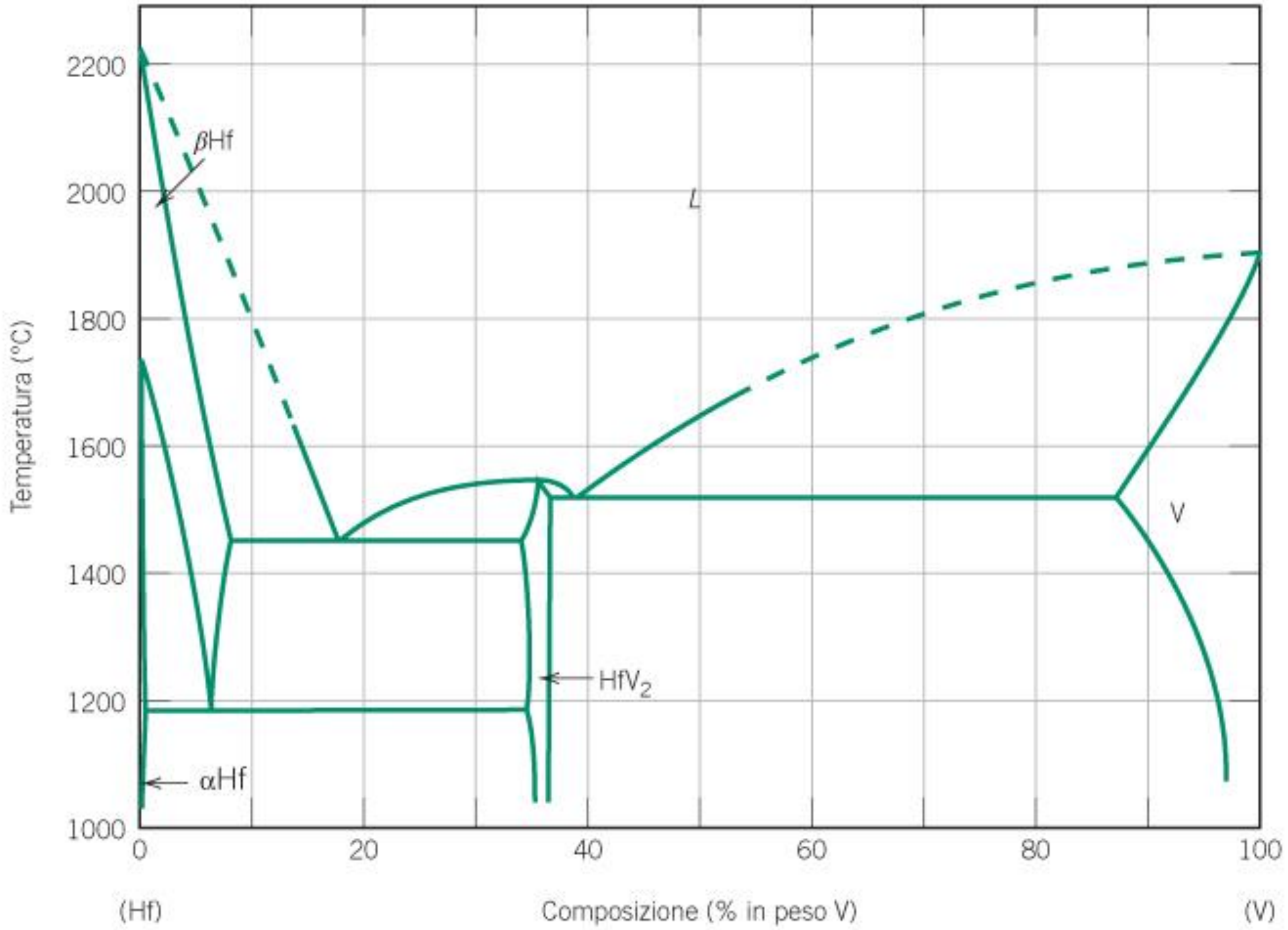
A constitution point in the two-phase field: liquid plus Pb-rich solid.

Find the proportions of each phase



Determine the phase/s present, the composition/s and relative amount/s for an alloy with 50%Ti at 1100°C.

Complete the following diagram



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Complete the following diagram

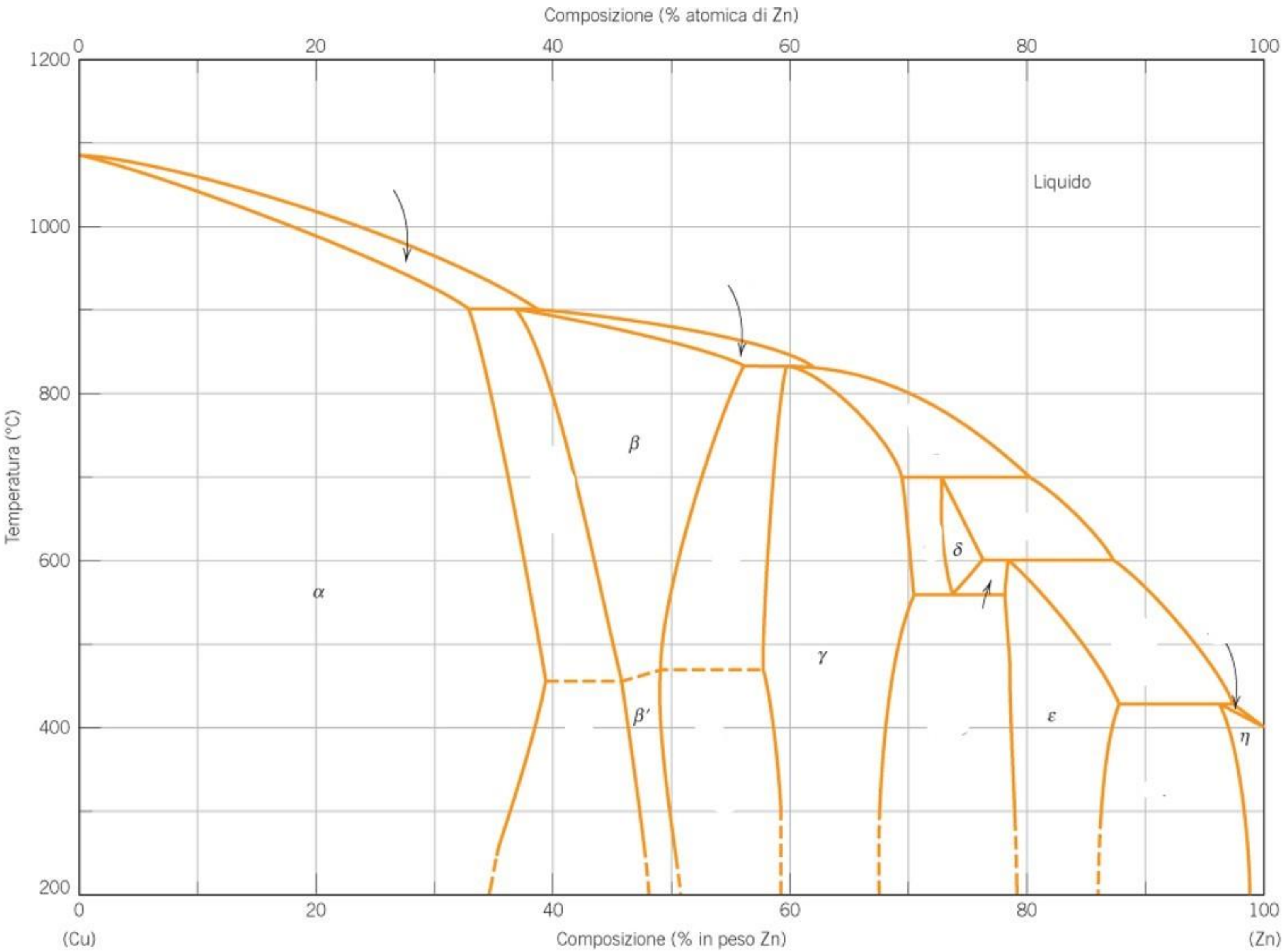

Figura 9.19 Diagramma di fase rame–zinco.

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